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Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-051630
Article Type:	Protocol
Date Submitted by the Author:	24-Mar-2021
Complete List of Authors:	Lu, Yue; Wuhan Sports University, Department of Health Sciences Xia, Yuan; Wuhan Sports University, Department of Health Sciences Li, Yongjie; Wuhan Sports University, Department of Health Sciences Pan, Xinyong; Wuhan Sports University, Department of Health Sciences Wang, Zhenyu; Wuhan Sports University, Department of Health Sciences
Keywords:	Stroke < NEUROLOGY, Neurology < INTERNAL MEDICINE, Rehabilitation medicine < INTERNAL MEDICINE, REHABILITATION MEDICINE

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Repetitive transcranial magnetic stimulation for upper limb motor function and activities of daily living in stroke patients: A protocol of a systematic review and Bayesian network meta-analysis

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"Yue Lu and Yuan Xia contributed equally to this paper"

Word count:2765

Keywords: repetitive transcranial magnetic stimulation; upper limb; motor function; activities of daily living; stroke; network meta-analysis

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Abstract

Introduction: Patients with stroke usually suffer from varying degrees of movement dysfunction, among which the movement dysfunction of the upper limb for seriously affecting the quality of life is evident.More and more studies have shown that repetitive transcranial magnetic stimulation (rTMS) has a positive therapeutic effect on upper limb movement function and activity of daily living of stroke patients. However, currently, there is a lack of effective comparison between efficacies of rTMS with different stimulation modalities. Therefore, this study aims to compare the effects of different rTMS modalities on the upper limb motor function and activities of daily living in stroke patients.

Methods and analysis: A systematic search will be conducted on PubMed, Web of Science, Embase, Cochrane Library, ProQuest, Wanfang Database , China National Knowledge Infrastructure（CNKI) and Chinese Scientific and Journal Database(VIP) for randomized controlled trials of rTMS in the treatment of stroke from their conception up to November 2020. Upper extremity Fugl-Meyer assessment scale(UE-FMA), Wolf motor function test (WMFT), Jebsen Taylor hand function test(JTHF), Action research arm test(ARAT), Box and block test(BBT) and other scales related to measuring of upper limb motor function will be performed as the primary outcomes. The secondary outcome measures include modified Barthel Index (MBI), which evaluates patients' activities of daily living and adverse events (such as vertigo, headache, epilepsy, etc.). To address personal bias in the inclusion of the study, the extraction of data and the quality evaluation will be carried out independently by the two reviewers. All data analyses will be performed by GeMTC0.14.3 and STATA16.0.

Ethics and dissemination: This Network Meta-Analysis (NMA) will not require ethical approval since the data analysis will be used only to evaluate the efficacy of rTMS treatment and will not involve patient private information. The results will be disseminated in international conference reports and peer-reviewed manuscripts.

PROSPERO registration number: CRD42020212253

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Strengths and limitations of this study

1 This is the first comprehensive Bayesian network meta-analysis to compare the effects of different rTMS modalities on the upper limb motor function and activities of daily living in stroke patients.

2 The results from this network meta-analysis will find out the best stimulation mode, and provide preliminary evidence for clinical treatment of stroke patients.

3 Due to the uneven methodological quality of the literature included, there may be lead to a bias in the results.

4 We will systematically search Chinese and English databases, while other languages may be ignored.

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Introduction

Stroke is a common disease that seriously endangers human health, characterized by high incidence, high morbidity, high recurrence and high mortality rate, laying heavy economic burden on patients and their families. It is a major cause of long-term disability in the world [1]. Stroke is often accompanied by abnormal posture control, impaired balance function, abnormal muscle strength and other symptoms, which may trigger motor dysfunction with various degrees of severity. Studies [2] have shown that about 55% and 75% of stroke patients suffer from upper limb motor dysfunction. The limitation of the upper limb often leads to the limited activity of daily living and the low quality of life [3]. Thus, the research hotspot in finding a effective and safe way to improve the limb motor function and activity of daily living of stroke patients .

Repetitive transcranial magnetic stimulation (rTMS) is a neuroelectrophysiological technique, using a certain intensity of time-varying magnetic field to generate an induced electric field in the brain, which can depolarize neurons and change the excitability of the local cortex [4]. According to the stimulation mode, rTMS can be divided into single-pulse transcranial magnetic stimulation and paired-pulse transcranial magnetic stimulation [5]. Single-pulse transcranial magnetic stimulation can be divided into low frequency 1HZ (≦1HZ) and high frequency stimulation (>1HZ) [6]. According to different delivery modes, the paired-pulse transcranial magnetic stimulation can be divided into Continuous Theta Burst Stimulation (CTBS) and Intermittent Theta Burst Stimulation (ITBS). The parameters affecting rTMS include stimulation site, frequency, pulse number and so on, chief among these is the frequency [7]. It is generally believed that low-frequency（≦1HZ）can suppress the excitability of the cerebral motor cortex and cause long-term synaptic inhibition, while high-frequency （≧5HZ） can facilitate and excite the cerebral cortex [8 9]. However, this is may not be applicable to all cases, and the specific effect also depends on the state of brain activity at the site of stimulation [10]. Previous studies have shown that after stroke [11], the balance of mutual inhibition between the two cerebral hemispheres is broken, that is, the excitability of the affected hemispheres decreases while the excitability of the healthy hemispheres increases, and the

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inhibition of the healthy hemispheres on the affected hemispheres strengthens, thus affecting the recovery of the upper limb function of the patients. In recent years, there are related studies [12-14] to observe the effects of the above stimulation on upper limb movement function and activities of daily living in stroke patients, but these results are still controversial and lack effective comparison between different stimulation modalities. Therefore, this research attempts to compare the effects of rTMS on upper limb movement function and activities of daily living in stroke patients through network meta-analysis which studies direct and indirect evidence, and to find out the most effective and safe rTMS parameters and to used in the clinical practice.

Methods

The protocol of this systematic review will be developed and reported on the basis of the preferred reporting project of the system review and meta-analysis[15] (PRISMA-P) guidelines, and has been registered in the international system evaluation expected register (PROSPERO, registration number: CRD42020212253). We will implement the research protocol in accordance with the preferred reporting project guide for system evaluation and meta-analysis.

Inclusion criteria

Types of studies

We will evaluate the publication against the criteria of subjects, interventions, controls, and outcome (PICO). This study will only involve randomized controlled trials of the effects of rTMS on upper limb movement function and activities of daily living in patients with stroke, and focus on studies written in Chinese and English. In addition, incorrect randomized methods, non-randomized controlled trials, case reports and other studies will be excluded.

Types of participants

Only patients diagnosed with stroke will be included. All subjects must meet the diagnostic criteria of stroke in their countries and are further diagnosed with stroke by

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CT or MRI. They are supposed to be over eighteen years old with clear consciousness and no cognitive impairment, regardless of their sexes and courses of disease.

Types of interventions

Stimulation groups will be divided into three subgroups, the low frequency rTMS one, the high frequency rTMS one and TBS one. The control group will include routine treatment group and sham stimulation group. Routine treatment group will received routine rehabilitation treatment, including OT, PT, VR, orthosis, etc. Sham stimulation group will simulate the sound of real stimulation during treatment, but will not produce effective magnetic stimulation. Studies on the combination of drugs that relieve spasms (such as botulinum toxin, etc) will be excluded.

Outcome measurements

The primary outcomes will focus on the motor function of the upper limb, while the secondary outcomes will include activities of daily living and adverse reactions.

Primary outcomes

The main outcome indicators will be Upper extremity Fugl-Meyer assessment scale(UE-FMA), Wolf motor function test (WMFT), Jebsen Taylor hand function test(JTHF), Action research arm test(ARAT), Box and block test(BBT) and other scales related to measuring of upper limb motor function.UE-FMA consists of 33 items evaluating motor function of patients' upper limbs [16], each of which is scored by 0, 1 or 2. A score of 0, 1 and 2 represents complete activity limitation, partial activity limitation and no activity limitation, respectively. The total score is 66, and higher scores indicates better upper limb motor function in patients. FMA contains: upper limb reflex activity; extensor and flexor joint movement; joint movement; dissociation movement; normal reflex activity; stability of the shoulder, elbow and wrist; motor function of fingers as well as coordination and speed. WMFT is used to evaluate the function of hands in performing task activities [17], with a total of 15 items, including 6 items to evaluate the motor function of upper limbs and 9 items hands to complete functional tasks. During the test, the time of completion of the patient's action is

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recorded and the quality of the completed action is scored with a total of 6 grades of 0-5. The higher score means the higher quality of the action completed by the patient will be. JTHF mainly evaluates the speed of upper limb movement [18]. It has seven items: writing, flipping cards, picking up small items into containers, imitating eating, stacking chess pieces, moving large and light objects and moving large and heavy items. Record the time it takes to complete each activity, and the maximum time allowed for each activity is 180 seconds. The BBT measures the number of one-inch building blocks that subjects move from one box to another in a minute, and the more the number, the better their athletic ability [19]. ARAT is designed to evaluate the motor ability of the upper limb after stroke, which consists of four parts: grasp, grip, pinch and gross movement. There are 19 items, with 4 grades in each item, and the highest aggregate score is 57. The higher score indicates the better motor ability [20].

Secondary outcomes

Secondary outcome indicators will include modified Barthel Index (MBI) and adverse events.MBI is used to measure the activities of daily living of patients [21] , including defecation control, eating, dressing, going upstairs and downstairs, going to the toilet, transferring, walking, bathing and grooming. The full score was 100, and Higher score represents better ability of daily living. Adverse events include vertigo, epilepsy, headache, sensory abnormalities and so on.

Exclusion criteria

The following contents were excluded :(1) repeated publication of literature;(2) the following study types: literature review, cross-over study , cohort study, or case-control study;(3) data that cannot be extracted;(4) upper limb motor dysfunction is not caused by stroke(e.g., trauma, cerebral palsy, Parkinson, etc.) ;(5) inaccessible texts.

Search strategies

We will systematically search the PubMed, Embase, Cochrane Library, Web of Science, ProQuest, China National Knowledge Infrastructure(CNKI), Wanfang Database and Chinese Scientific and Journal Database(VIP) for the randomized

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controlled trials of patients with stroke treated by repetitive transcranial magnetic stimulation. The search time from inception up until November 2020, and the retrieval strategy will be the method of subject words combined with free words. Key words will be included: stroke, rTMS, transcranial magnetic stimulation, repetitive transcranial magnetic stimulation, etc.. Specific search (take Pubmed) as an example: (Stroke [mesh] OR cerebrovascular accident [title/Abstract] OR CVA [Title/Abstract] OR Brain Vascular Accident [Title/Abstract] OR hemiplegia [Title/Abstract] OR (apoplexy [Title/Abstract] OR (hemiparesis [Title/Abstract]) AND (repetitive transcranial magnetic stimulation [Title/Abstract] OR Transcranial Magnetic proposal [title/Abstract] OR TMS [Title/Abstract] OR rTMS [Title/Abstract] OR Theta burst stimulation [Title/Abstract] OR θ burst stimulation [Title/Abstract]). The Chinese database will be searched with Chinese characters in the same meaning.

Studies selection

After removing all the repeating documents, two reviewers (YL and YX) will independently selected the literature. According to the above inclusion criteria, the two reviewers will make the first selection by reading the title and abstract of the literature, and then downloaded the full text. Then the two reviewers will further pick out literature that meet the inclusion criteria by reading the full text. We will contact the author if the selected literature had incomplete information or data. Any disagreements in the inclusion process will be handled through a group discussion or consultation with an experienced reviewer (YJL). The study of and the reasons for the exclusion will be recorded, and the details of the entire literature screening process will be shown in the PRISMA flowchart [22], as shown in figure 1.

Data extraction

The literature data will be extracted by two reviewers (XP and ZW) using previously designed tables to extract data from each study. The extracted data will be included: first author and year of publication, country of publication, course of the disease, sample size, age, mode of intervention, duration of intervention, stimulation

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site, and outcome measurements. The inclusion data will be collected by Excel, and the two reviewers will cross-check the inclusion data. As For the missing data information, we will contact the author for further information.In addition, if there was any disagreement, the research group discussed and resolved it with a mutual consensus.

Risk of bias assessment

Two reviewers (Xia and Lu) will independently assess the bias risk of the included study by using the Cochrane collaborative bias risk tool [23]. The evaluation will include random sequence generation, distribution hiding, blinding subjects and therapists, blinding the outcome evaluation, incomplete outcome data, selective reporting results and other biases. The reviewers will assess the bias risk included in the study as low, unclear and high risk in accordance with the methodology of the Cochrane intervention system Review Manual. If encounter any differences, the research group will jointly discuss and resolve them. Finally, the bias risk diagram will be performed on Revmen5.3.

Grading the quality of evidence

The two reviewers will use the GRADE system to independently evaluate the quality of the evidence of the results. The GRADE system will include the following five aspects, studying the limitations, inconsistencies, indirectness, inaccuracies and publication bias of the design[24]. The quality of the evidence will be labeled as "high", "medium", "low" or "very low" according to the rating criteria[25]. An evidentiary overview of the rating results and a table of findings will be generated using GRADE PRO software.

Dealing with missing data

If the data in this study is missing or the information is unclear, the first author or the corresponding author will first be contacted to obtain the relevant data. If no reply is received, we will try to calculate the data through the available coefficients. In

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addition, if the data is reported in the form of a picture, the data will be extracted through GetdataV2.25. The potential impact of missing data on the results will be explained in the discussion.

Ethics and dissemination

Since this study involves no patient privacy, there is no need for ethical assessment. The data will be processed in accordance with the agreement, the systematic review will be published in peer-reviewed journals or conference reports, and the analysis will be conducted in accordance with PRISMA guidelines.

Statistical analysis

Stata16.0 will be used to create network diagrams of treatments for patients with stroke and a comparison-adjusted funnel plot for network meta-analysis. Dots in the mesh map represent different interventions, and the size of dots represents the sample size included in this intervention; the connection between the dots represents a direct comparison between the two interventions, and there is no direct comparison between the two interventions. The thickness of the lines represents the number of studies included between the two interventions. The comparison-adjusted funnel plots will be used to evaluate whether there is a small sample effect or publication bias in the included study.

In addition, we will use GeMTC0.14.3 software (Generate Mixed Treatment comparison) and MarkovChain Monte Carlo(MCMC) to carry out Bayesian inference. Because the outcome measurements involved in this study are mainly continuous variables, the mean difference and 95%CI will be used as the effect size. The parameters for GeMTC will be set as follows: the simulation chain is set to 3, the step (thinning interval) is set to 10, and the number of iterations is 50000, and the first 20000 are used for annealing to eliminate the effect of the initial value. The node splitting method is used to test the inconsistency between direct evidence and indirect evidence. When $P > 0.05$, the inconsistency is not obvious, so the consistency model could be used for data analysis. The convergence between the included studies is

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expressed by the potential scale reduction parameter (Potential Scale Reduced Factor,PSRF). When the PSRF is close to or equal to 1, it will mean the convergence is good, and the results with high reliability can be obtained by using the consistency model analysis. Evaluating the curative effect of each intervention method by drawing the probability ranking plot. If the number of studies included is sufficient, subgroup analyses will be performed considering the duration of onset(≦6 months and ≧6 months), the duration of intervention and the type of stroke(ischemic and hemorrhagic).

Sensitivity analysis

In order to assess the stability of the results, a sensitivity analysis of the preliminary results will be conducted to assess the impact of methodological quality, research quality, sample size and missing data in this review.

Publication bias

We will use the Egger's test and funnel chart to assess the publication bias of the included major findings. If the funnel diagram is found asymmetrical, we will try to explain the asymmetry of the funnel diagram.

Discussion

Patients with stroke are usually accompanied by a variety of dysfunction, among which the dysfunction of the upper limb is more frequent, and the dysfunction of the upper limb seriously affects their quality of life[26]. Thus, the research hotspot in finding a effective and safe way to improve the limb motor function and activity of daily living of stroke patients . The excitability imbalance of the cerebral hemisphere after stroke is characterized by the abnormal increase of cortical activity on the healthy side[27] [28], while the cerebral activity on the affected side decreased significantly due to the interhemispheric inhibition of the contralateral hemisphere [29] [30],which seriously impedes the recovery of motor function of the patients. Repetitive transcranial magnetic stimulation is a non-invasive and relatively safe

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neuroelectrophysiological technique, which can affect the neuroelectrical activity of the cerebral cortex and regulate the plasticity of the brain. It has been widely used in the field of post-stroke rehabilitation[31]. Low frequency repetitive transcranial magnetic stimulation (LF-rTMS) and continuous theta burst stimulation (CTBS) can inhibit cortical excitability and lead to motor cortex inhibition, while high frequency repetitive transcranial magnetic stimulation (HF-rTMS) and intermittent theta burst stimulation(ITBS) can increase cortical excitability for a long time[32 33]. Previous meta-analysis[34 35] shows that rTMS plays a positive role in improving upper limb movement and activities of daily living in stroke patients, but the above evidence is obtained by direct comparison (that is, each stimulation group is compared with the control group and the sham stimulation group), while a systematic comparison between different stimulation modalities is lacking. This network meta-analysis will include and integrate the latest and most comprehensive literature in this field. Our aim is to compare the effects of repetitive transcranial magnetic stimulation with different stimulation modalities on upper limb movement function and activities of daily living in stroke patients by network meta-analysis, and then find out the best stimulation mode, accordingly provide preliminary evidence for clinical treatment of stroke patients.

Contributors YL,YX and YjL conceived the content and drafted the paper. XP and ZW developed the search strategies and conducted data collection. YjL approved the final version of the manuscrip.All authors have approved the final manuscript.

Funding This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests None declared.

Patient and public involvement Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

Patient consent for publication Not required.

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Figure Legends

Figure 1 Flowchart of the study selection process

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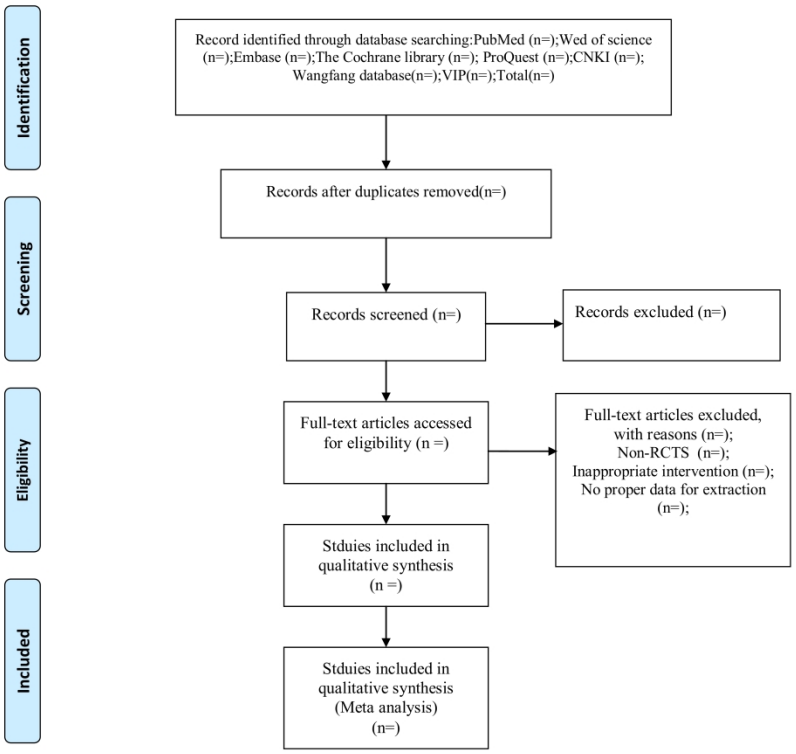


Figure 1 Flowchart of the study selection process

215x279mm (600 x 600 DPI)

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In your methods section, say that you used the PRISMA-Preporting guidelines, and cite them as:

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		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	title page
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	n/a

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Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	3
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	title page
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	12
Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	n/a
Support			
Sources	#5a	Indicate sources of financial or other support for the review	12
Sponsor	#5b	Provide name for the review funder and / or sponsor	12
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	12
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is	4

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		already known	
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	5
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	5-6
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	7
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	7-8
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	8
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	8
Study records - data collection	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any	8

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process		processes for obtaining and confirming data from investigators	
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	8-9
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	6-7
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	9
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	25
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	10
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	10
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	n/a
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within	10

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		studies)	
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	9

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BMJ Open

Repetitive transcranial magnetic stimulation for upper limb motor function and activities of daily living in stroke patients: A protocol of a systematic review and Bayesian network meta-analysis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-051630.R1
Article Type:	Protocol
Date Submitted by the Author:	16-Dec-2021
Complete List of Authors:	Lu, Yue; Wuhan Sports University, Department of Health Sciences Xia, Yuan; Wuhan Sports University, Department of Health Sciences Wu, Yue; Wuhan Sports University Pan, Xinyong; Wuhan Sports University, Department of Health Sciences Wang, Zhenyu; Wuhan Sports University, Department of Health Sciences Li, Yongjie; Guizhou Orthopedics Hospital, Department of rehabilitation medicine
Primary Subject Heading:	Evidence based practice
Secondary Subject Heading:	Rehabilitation medicine, Mental health
Keywords:	Stroke < NEUROLOGY, Neurology < INTERNAL MEDICINE, Rehabilitation medicine < INTERNAL MEDICINE, REHABILITATION MEDICINE

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Repetitive transcranial magnetic stimulation for upper limb motor function and activities of daily living in stroke patients: A protocol of a systematic review and Bayesian network meta-analysis

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Word count:3104

Keywords:repetitive transcranial magnetic stimulation;upper limb;motor function; activities of daily living;stroke;network meta-analysis

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Abstract

Introduction: Patients with stroke usually suffer from varying degrees of movement dysfunction,which seriously affects their quality of life, especially for the upper limb dysfunction.Therefore, this study aims to compare the effects of different rTMS modalities on stroke patients' upper limb motor function and daily activities.

Methods and analysis:Relevant research will be collected systematically from PubMed, Web of Science, Embase, Cochrane Library, ProQuest, Wanfang Database, China National Knowledge Infrastructure （ CNKI), and Chinese Scientific and Journal Database(VIP) about randomized controlled trials of rTMS in the stroke treatment range from the establishment to November 2020. Primary outcomes will be obtained from scales measuring the upper limb motor function like Upper extremity Fugl-Meyer assessment scale(UE-FMA), Wolf motor function test (WMFT), JebsenTaylorhandfunction test(JTHF), Action research arm test(ARAT)and Box and block test(BBT) .The secondary outcomes include modified Barthel Index (MBI) and adverse events (such as vertigo, headache, and epilepsy), with the goal of assessing patients' activities of daily living and the safety of treatment. In order to avoid personal bias in the included studies, two reviewers will conduct the data extraction and quality evaluation independently, and all data analyses will be performed by GeMTC0.14.3 and STATA16.0.

Ethics and dissemination: The Network Meta-Analysis (NMA) in this study does not require ethical approval because the data analysis will be used only to evaluate the rTMS treatment efficacy without patients' private information. In addition, the results will be disseminated in international conference reports and peer-reviewed manuscripts.

PROSPERO registration number: CRD42020212253

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Strengths and limitations of this study

1This is the firstcomprehensive Bayesian network meta-analysistocompare the effects of different rTMS modalities on the upper limb motor function and activities of daily livinginstroke patients.

2 The network meta-analysis contributes to findingout the best stimulation modeand provides preliminary evidence for clinical treatment of stroke patients.

3 Bias may exist in the results due to the uneven methodological quality of the literature.

4 We will systematically search bothChinese and Englishdatabases.

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Introduction

As a common disease that seriously threatens human health, stroke is characterized by high incidence, high morbidity, high recurrence, and high mortality rate, laying heavy economic burdens on patients and their families. It is a major cause of long-term disability in the world.[1] Stroke is often accompanied by abnormal posture control, impaired balance function, abnormal muscle strength, and motor dysfunction. Studies show that about 55% and 75% of stroke patients suffer from upper limb motor dysfunction, limiting their daily activities and negatively affecting their life.[2-4] Thus, this study wants to put forward an effective and safe way to improve stroke patients' limb motor function and activity of daily living.

Repetitive transcranial magnetic stimulation (rTMS) is a neuroelectrophysiological technique, using a time-varying magnetic field with certain intensity to generate an induced electric field in the brain, depolarizing neurons and changing the excitability of the local cortex.[5] According to the frequency, conventional rTMS can be divided into low-frequency stimulation (≦ 1HZ) and high-frequency stimulation (>1HZ).[6, 7] Currently,theta-burst stimulation (TBS) has received widespread attention as an emerging patterned rTMS (prTMS).[8] Based on the difference between stimulation and intermittent time, and the effect on the excitability of the cerebral cortex, TBS can be divided into intermittent TBS (ITBS) with excitatory effects and continuous TBS(CTBS) producing inhibitory effects.[9] The parameters affecting rTMS include stimulation site, frequency, pulse number,etc., among which frequency is the primary factor.[10] Generally, low-frequency(≦1HZ)is believed to suppress the excitability of the cerebral motor cortex and cause long-term synaptic inhibition, while high-frequency(≧5HZ) can facilitate and excite the cerebral cortex.[11, 12] However, this may not apply to all cases, and the specific effect depends on brain activity at the site of stimulation.[13] According to previous studies, the balance of mutual inhibition between the two cerebral hemispheres is broken after stroke, which means that the excitability of affected hemispheres decreases while that of healthy hemispheres increases.[14] Meanwhile,the inhibition of healthy hemispheres on affected hemispheres

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strengthens, thus affecting the recovery of the upper limb function. Besides, there are also theories suggesting that the activation of the contralateral hemisphere through high-frequency stimulation can promote the reorganization of brain function and enhance compensatory ability, thereby improving the motor function of stroke patients.[15, 16]

In recent years, there have been related studies observing the effects of the above stimulation on upper limb motor function and activities of daily livingin stroke patients.[17, 18] Most evidence were obtained by comparison with routine rehabilitation treatment, while either direct comparionsbetween different modalities of rTMS are lacking. In addition, some previous traditional meta-analyses have revealed the effectiveness of rTMS on stroke patients.[19-21] However, limited by the availability of pairwise comparisons between interventions, it is difficult to draw a conclusion about which is the most comparatively effective and safe stimulation modalities. In this case, network meta-analysis can provide direct and indirect comparisons among multiple interventions.[22] Therefore, this study attempts to compare the effects of different modalities of rTMS on upper limb motor function and activities of daily living in stroke patients through network meta-analysis with both direct and indirect evidence.The first research questionis: what isthe most effective rTMS modality on upper limb movement function and activities of daily livingin stroke patients? This is the most important result, which is helpful for clinical selection of the best stimulation modality. The second research question is: How safe are the different modalities of rTMS? Investigating the potential adverse reactions of different stimulation modalities is conducive to the standardized use of rTMS.

Methods

The protocol of this systematic review will be developed and reported based on the preferred reporting project of the system review and meta-analysis(PRISMA-P) guidelines[23] and has been registered in the international system evaluation expected to register (PROSPERO, registration number: CRD42020212253). The research protocol will follow the preferred reporting project guide for system evaluation and

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meta-analysis.

Inclusion criteria

Types of studies

The publication against the criteria of subjects, interventions, controls, and outcome (PICO) will be evaluated.This study will only involve randomized controlled trials of the effects of rTMS on upper limb motor function and activities of daily living in patients with stroke and focus on studies written in Chinese and English. In addition, incorrect randomized methods, non-randomized controlled trials, case reports, and other studies will be excluded.

Types of participants

Only patients diagnosed with stroke will be chosen as participants. All subjects must meet the diagnostic criteria of stroke and should be further clinically diagnosed withstroke by CT or MRI. Additionally, they should be over eighteen years old with clear consciousness and no cognitive impairment, regardless of their sexes and courses of disease.

Types of interventions

Subgroups are included in stimulation groups: the low frequency rTMS (LF-rTMS) on unaffected side, the high frequency rTMS (HF-rTMS) on affected side, the high frequency rTMS on unaffectedside, ITBS and CTBS. The control group will include the routine treatment group and sham stimulation group. The routine treatment group will receive routine rehabilitation treatment, including occupational therapy (OT), physical therapy (PT), exercise therapy, virtual reality technology (VR), orthotics, etc. As for the Sham stimulation group, the sound of real stimulation will be simulated during treatmentwithoutproducingeffective magnetic stimulation. Studies on the combination of drugs that relieve spasticity (such as botulinum toxin) will be excluded.

Outcome measurements

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The primary outcomes will focus on the motor function of the upper limb, while the secondary outcomes will discussactivities of daily living and adverse reactions of stroke patients.

Primary outcomes

The main outcome indicators will bethe Upper Extremity Fugl-Meyer assessment scale(UE-FMA), Wolf motor function test (WMFT), JebsenTaylor hand function test(JTHF), Action research arm test(ARAT), Box and block test(BBT), and other scales to measure the upper limb motor function.UE-FMA consists of 33 items evaluating the motor function of patients' upper limbs, eachscored by 0, 1, or 2. A score of 0, 1, and 2 represents complete activity limitation, partial activity limitation, and no activity limitation, respectively.[24] Higher scores indicate better upper limb motor function in patients, and the total score is 66. FMA contains upper limb reflex activity, extensor, and flexor joint movement, joint movement, dissociation movement, normal reflex activity, stability of the shoulder, elbow, and wrist, motor function of fingers, coordination, and speed. WMFT is used to evaluate the function of hands in performing task activities with 15 items, including 6 items to evaluate the motor function of upper limbs and 9 items to complete functional tasks.[25] During the test, the time of the patient's action is recorded, and the performance of actions will be scored with a total of 6 grades of 0-5. A higher score corresponds to a better quality of the action completed by the patient. In addition, JTHF mainly evaluates the speed of upper limb movement.[26] It involves seven items: writing, flipping cards, picking up small items into containers, imitating eating, stacking chess pieces, moving large and light objects, and moving large and heavy items. The time to complete each activity will be recorded, and the maximum time allowed for each activity is 180 seconds. BBT measures the number of one-inch building blocks that subjects move from one Box to another in a minute, and the larger number reflects better athletic ability.[27] ARAT is designed to evaluate the motor ability of the upper limb after stroke, which consists of four parts: grasp, grip, pinch, and gross movement. There are 19 items with each of 4 grades, and the highest aggregate score is 57. Similarly, a higher score indicates better motor ability.[28]

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Secondary outcomes

Secondary outcome indicators will include modified Barthel Index (MBI) and adverse events.MBI is used to measure the activities of daily living of patients, including defecation control, eating, dressing, going upstairs and downstairs, going to the toilet, transferring, walking, bathing, and grooming.[29] The full score is100, and the higher score represents a better ability of daily living. Adverse events include vertigo, epilepsy, headache, sensory abnormalities, etc.

Exclusion criteria

The following contents were excluded:(1) repeated publication of literature;(2) literature review, cross-over study, cohort study, or case-control study;(3) data that cannot be extracted;(4) upper limb motor dysfunction not caused by stroke(e.g., trauma, cerebral palsy, and Parkinson);(5) inaccessible texts;(6)The rTMS stimulation site except the M1 area.

Search strategies

We will systematically search the PubMed, Embase, Cochrane Library, Web of Science, ProQuest, China National Knowledge Infrastructure(CNKI), Wanfang Database and Chinese Scientific and Journal Database(VIP) for randomized controlled trials of stroke patients treated by repetitive transcranial magnetic stimulation. The time range will be from inception to November 2020, and the retrieval strategy will be subject words combined with free words. Keywords of stroke, rTMS, transcranial magnetic stimulation, repetitive transcranial magnetic stimulation will be includedin the search. Specific search (take Pubmedas an example): (Stroke [mesh] OR cerebrovascular accident [title/Abstract] OR CVA [Title/Abstract] OR Brain Vascular Accident [Title/Abstract] OR hemiplegia [Title/Abstract] OR (apoplexy [Title/Abstract] OR (hemiparesis [Title/Abstract]) AND (repetitive transcranial magnetic stimulation [Title/Abstract] OR Transcranial Magnetic proposal [title/Abstract] OR TMS [Title/Abstract] OR rTMS [Title/Abstract] OR Theta burst stimulation [Title/Abstract] OR θ burst stimulation [Title/Abstract]). The Chinese

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database will be searched with Chinese counterpartsin the same approach.

Studies selection

After removing all the repeating documents, two reviewers (YL and YX) will independently select the literature. The two reviewers will select the above inclusion criteria by reading the literature's title and abstract before downloading the full text. Afterward, the two reviewers will pick out literature that satisfies the inclusion criteriaby reading the full text. We will contact the author if the selected literature fails to provide complete information or data. Any disagreements in the inclusion process will be handled through a group discussion or consultation with an experienced reviewer (YJL). Reasons for the exclusion will be recorded, and the details of the entire literature screening process will be presentedin the PRISMA flowchart, [30] as shown in figure 1.

Data extraction

Two reviewers (XP and ZW) will extract the literature data using previously designed tables from each study. The extracted data containthefirst author,year of publication, country of publication, disease course, sample size, age, mode of intervention, duration of intervention, stimulation site, and outcome measurements. The inclusion data will be collected by Excel, which the two reviewers will cross-check. As for the missing data, we will contact the author for completeinformation.In addition, in the case of any disagreement , the research group will discuss and resolve with mutual consensus.

Risk of bias assessment

Two reviewers (Xia and Lu) will independently assess the bias risk of the included study using the Cochrane collaborative bias risk tool.[31] The evaluation will include random sequence generation, distribution hiding, blinding subjects and therapists, blinding outcome evaluation, incomplete outcome data, selective reporting results, and other biases. The reviewers will assess the bias risk included in the study as low,

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unclear, and high risk under the Cochrane intervention system Review Manual. The research group will jointly discuss and resolve any disagreement. Finally, the bias risk diagram will be performed on Revmen5.3.

Grading the quality of evidence

The two reviewers will use the GRADE system to evaluate the quality of the evidence of the results independently. The GRADE system will include the following five aspects: the limitations, inconsistencies, indirectness, inaccuracies, and publication bias of the design.[32] The quality of the evidence will be labeled as "high", "medium", "low" or "very low" according to the rating criteria.[33] An evidentiary overview of the rating results and a table of findings will be generated via GRADE PRO software.

Dealing with missing data

If the data is missing or unclear, the first author or the corresponding author will be contacted to obtain relevant information. If no reply is received, we will try to calculate the data through the available coefficients. In addition, if the data is reported in the form of a picture, the data will be extracted through GetdataV2.25. The potential impact of missing data on the results will be explained in the discussion.

Statistical analysis

Stata16.0 will create network diagrams of treatments for stroke patients and a comparison-adjusted funnel plot for network meta-analysis. Dots in the mesh map represent different interventions, and the size of dotsrepresents the sample size; the connection between the dots indicates a direct comparison, and no direct comparison is observed between the two interventions. In addition, the thickness of lines represents the number of studies included.The comparison-adjusted funnel plots will be used to evaluate whether there is a sample effect or publication bias in the included study.

Moreover, we will adopt GeMTC 0.14.3 software (Generate Mixed Treatment

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comparison) and Markov Chain Monte Carlo (MCMC) to carry out Bayesian inference. Considering thatthe outcome measurements involved in this study are mainly continuous variables, the mean difference and 95%CI will be used as the effect size. The parameters for GeMTC will be set as follows: 3 simulation chains, 10 steps (thinning interval), 50000 iterations, and the first 20000 are used for annealing to eliminate the influenceof the initial value. The node splitting method is used to test the inconsistency between direct and indirect evidence. When $P>$ 0.05, the inconsistency is not obvious, enabling the consistency model for data analysis. The convergence between the included studies is expressed by the potential scale reduction parameter (Potential Scale Reduced Factor, PSRF). When the PSRF is close to or equals1, the convergence should begood. The results with high reliability can be obtained via the consistency model analysis, and the probability ranking plot will help evaluate the curative effect of each intervention method. If the number of included studiesis sufficient, subgroup analyses will be performed considering the duration of onset(≦6 months and ＞6 months), the duration of intervention, the type of stroke(ischemic and hemorrhagic),the location of onset(basal ganglia, thalamus, brainstem, cerebellum, etc.),and extent of stroke(we will use half of the UE-FMA score as the boundary between mild and severe stroke.UE-FMA score<33:severe stroke; UE-FMA score≧33: mild stroke).

Sensitivity analysis

In order to verify the stability of the results, a sensitivity analysis of the preliminary results will be conducted to evaluate the impact of methodological quality, sample size, and missing data on pooled results of this study.

Publication bias

We will use Egger's test and funnel chart to assess the publication bias of the included major findings. If the funnel diagram is asymmetrical, we will explain the asymmetry of the funnel diagram.

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Patient and public involvement

Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

Discussion

Patients with stroke are usually inflictedby dysfunction, and that of the upper limb is frequent and seriously affects their quality of life.[34] Thus, the research hotspot in finding an effective and safe way to improve stroke patients' limb motor function and daily activity. The excitability imbalance of the cerebral hemisphere after stroke is characterized by the abnormal increase of cortical activity on the unaffected side.[35, 36] In contrast, the cerebral activity on the affected side will decrease significantly due to the interhemispheric inhibition of the contralateral hemisphere, which seriously impedes the recovery of motor function of the patients.[37, 38] Repetitive transcranial magnetic stimulation is a non-invasive and relatively safe neuroelectrophysiological technique, which can affect the neuroelectrical activity of the cerebral cortex and regulate the plasticity of the brain. It has been widely used in post-stroke rehabilitation.[39] LF-rTMS on the unaffected side and CTBS can inhibit cortical excitability, while HF-rTMS on the affected side and ITBS can facilitate and increase cortical excitability.[8, 40] Additionally, there are also theories suggesting that the activation of the contralateral hemisphere through high-frequency stimulation can promote the reorganization of brain function and enhancethecompensatory ability ofstroke patients, and this review will include related studies.[15,16] Previous meta-analysis showed that rTMS could improve the upper limb movement and activities of daily livingin stroke patients.[41,42] However, the relevant evidence is obtained by direct comparison (each stimulation group is compared with the control group and the sham stimulation group) instead of systematically comparing different stimulation modalities.

This network meta-analysis will include and integrate the latest and most comprehensive literature in this field, aiming to compare the effects of repetitive

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transcranial magnetic stimulation with different stimulation modalities on the upper limb motor function and activities of daily livingin stroke patients by network meta-analysis. Then, this review intends to find out the best stimulation modeaccordingly andprovide preliminary evidence for the clinical treatment of stroke patients.

Ethics and dissemination

The Network Meta-Analysis (NMA) in this study does not require ethical approval because the data analysis will be used only to evaluate the rTMS treatment efficacy withoutpatients' private information. In addition, the results will be disseminated in international conference reports and peer-reviewed manuscripts.

ContributorsYL,YX and YjL conceived the content and drafted the paper.XP and ZW developed the search strategies and conducted data collection.YW provided important suggestions for the revision of the manuscript.YjL approved the final version of the manuscrip.All authors have approved the final manuscript.

FundingThis research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests None declared.

Patient consent for publication Not required.

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Figure Legends

Figure 1 Flowchart of the study selection process

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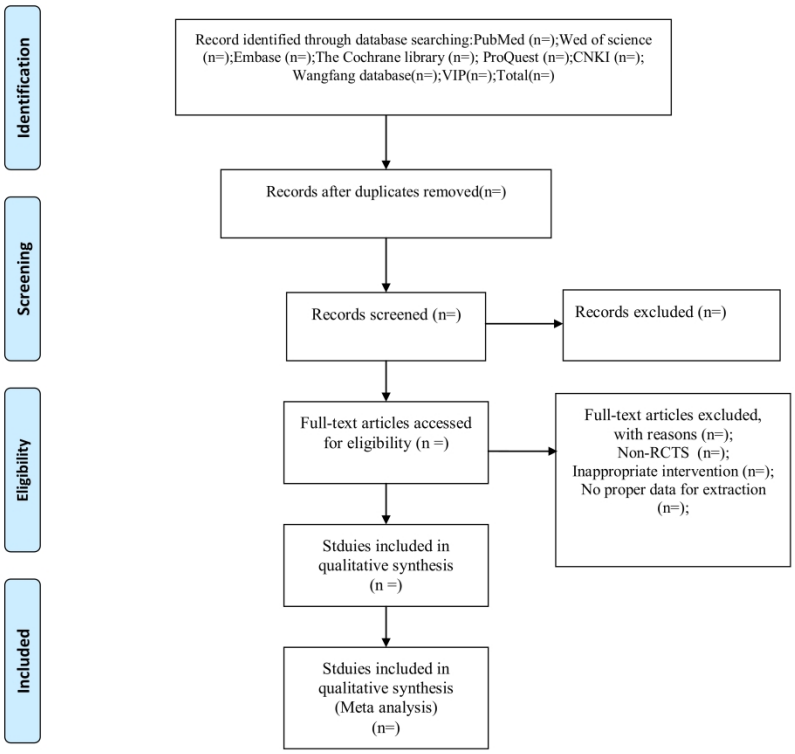


Figure 1 Flowchart of the study selection process

215x279mm (600 x 600 DPI)

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Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	title page
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	13
Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	n/a
Support			
Sources	#5a	Indicate sources of financial or other support for the review	13
Sponsor	#5b	Provide name for the review funder and / or sponsor	13
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	13
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is	5

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		already known	
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	5
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	6-8
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	8
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	8
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	9
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	9
Study records - data collection	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any	9

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process		processes for obtaining and confirming data from investigators	
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	6-8
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	7-8
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	9
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	10-11
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	10-11
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	11
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	n/a
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within	11

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		studies)	
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	10

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